At anode (oxidation): Iron dissolves in the anode region

$$2Fe(s) \rightarrow 2Fe^{2+}(aq)+4e^{-} \qquad E^{\circ} = 0.44V.$$

The electrons move through the iron metal from the anode to the cathode area where the oxygen dissolved in water, is reduced to water.

At Cathode (reduction)

The reaction of atmospheric carbon dioxide with water gives carbonic acid which furnishes the H^{+} ions for reduction.

$$O_2(g)+4H^{+}(aq)+4e^{-} \rightarrow 2H_2O\,(l) \qquad E^{\circ} = 1.23V$$

The electrical circuit is completed by the migration of ions through water droplet.

The overall redox reactions is,

$$2Fe(s)+O_2(g)+4H^{+}(aq) \rightarrow 2Fe^{2+}(aq) + 2H_2O(l) \qquad E^{\circ} = 0.444+1.23 = 1.67V$$

The positive emf value indicates that the reaction is spontaneous.

Fe^{2+} ions are further oxidised to Fe^{3+}, which on further reaction with oxygen form rust.

$$4Fe^{2+}(aq)+O_2(g)+4H^{+}(aq) \rightarrow 4Fe^{3+}(aq)+2H_2O(l)$$

$$2Fe^{3+}(aq)+4H_2O(l) \rightarrow Fe_2O_3.H_2O(s) + 6H^{+}(aq)$$

Other metals such as aluminium, copper and silver also undergo corrosion, but at a slower rate than iron. For example, let us consider the reduction of aluminium,

$$Al(s) \rightarrow Al^{3+}(aq)+3e^{-}$$

Al^{3+}, which reacts with oxygen in air to forms a protective coating of Al_2O_3. This coating act as a protective film for the inner surface. So,further corrosion is prevented.

Protection of metals form corrosion

This can be achieved by the following methods.

i. Coating metal surface by paint.

ii. Galvanizing - by coating with another metal such as zinc. zinc is stronger reducing agent than iron and hence it can be more easily corroded than iron. i.e., instead of iron, the zinc is oxidised.

iii. Cathodic protection - In this technique, unlike galvanising the entire surface of the metal to be protected need not be covered with a protecting metal. Instead, metals such as Mg or zinc which is corroded more easily than iron can be used as a sacrificial anode and the iron material acts as a cathode. So iron is protected, but Mg or Zn is corroded.

Passivation - The metal is treated with strong oxidising agents such as concentrated HNO_3. As a result, a protective oxide layer is formed on the surface of metal.

Alloy formation - The oxidising tendency of iron can be reduced by forming its alloy with other more anodic metals.

Example, stainless steel - an alloy of Fe and Cr.

XII U9 Electro Chemistry.indd 61 11-01-2021 14:44:34